

Supplementary Material

Beyond Paper Counts: Measuring Representation, Entry, and Persistence at ICML and NeurIPS, 2018–2024

This supplement accompanies the EPJ Data Science Regular Article draft. It documents measurement variants, observation coverage, model diagnostics, and frozen-output provenance. All person counts and results are aggregate. Linked row-level author records are not included in the submission package.

S1 Frozen design and reporting hierarchy

The study separates three headline analyses and one diagnostic analysis:

- RQ1: production-adjusted country representation;
- RQ2: the annual trajectory of title-only prior-to-index portfolio distance;
- D1: a diagnostic association between distance and experienced-coauthor status at entry; and
- RQ3: experienced-coauthor associations with mutually exclusive exact +2 pathways.

D1 was moved below the headline RQs before final manuscript assembly because its role is to test a premise used to interpret RQ3. This change altered the manuscript label but not the frozen input, sample, model, or estimate.

Table S1 Frozen file-to-manuscript label mapping

Manuscript label	Frozen output label	Frozen file	Reason
RQ2 distance trajectory	rq1	outputs/epj_distance_based_v10/tables/rq1_year_distance_descriptive.csv	Frozen before manuscript RQ relabeling
Diagnostic D1	rq2	outputs/epj_distance_based_v10/tables/rq2_experienced_coauthor_contrasts.csv	Frozen before diagnostic downgrade
RQ3 exact +2 pathways	rq3	outputs/epj_distance_based_v10/tables/rq3_exact_plus2_contrasts.csv	Manuscript RQ3 retains the frozen third-analysis label

Machine-readable source: `table_s1_frozen_file_label_mapping.csv`.

S2 Geographic representation audit

S2.1 Primary quantities and two OpenAlex inputs

For group g and year t , the production-adjusted representation index is

$$PRI_{g,t} = \frac{\text{accepted-program country credit}_{g,t} / \text{all accepted-program country credit}_t}{\text{broad-AI work-country credit}_{g,t} / \text{all broad-AI work-country credit}_t}.$$

The primary manuscript table reports first-listed-author affiliation-country credits and the frozen OpenAlex country-aggregate denominator. The primary country aggregates were retrieved on 11 August 2026 with `primary_topic.subfield.id:1702, type:article|conference-paper, annual publication-year filters, and group_by=authorships.institutions.country_code`. The sum of the query-matched work counts in the 2018–2024 API metadata is 864,215. This count describes the query universe; the annual denominator for a country's production share is the sum of observed country credits, not the metadata work count.

A separately collected work-level dataset contains 863,752 unique OpenAlex work IDs under the same field and document-type filters. It supplies the work-level specifications and denominator-country missingness analysis. It is not the direct input to the frozen primary country-aggregate series. The two collection outputs must therefore be distinguished in sample-size statements and provenance records. Work-ID deduplication was applied to the work-level dataset; conceptual publication versions were not merged, focal-venue works were not excluded, and document types were not independently verified as peer reviewed.

The primary country-aggregate denominator uses full country counting. In that denominator, each broad-AI work receives one full credit for every distinct affiliation country observed across all authorships; repeated authors from one country do not add credits, and multinational works contribute once to every represented country. Annual production shares divide group credits by the sum of those work-country credits. The role-unmatched primary rule aligns the numerator with the first-listed-author entrant unit used by the person-history analyses while treating the denominator as an all-authorship field-production baseline; it is audited against role-matched and fractional alternatives. The underlying country groups are mutually exclusive: China; India; core Anglophone (Australia, Canada, Ireland, New Zealand, the United Kingdom, and the United States); and other non-core. The core-Anglophone set is deliberately narrow and frozen as an accounting category, not an exhaustive English-use classification; assigning a country to other non-core does not imply low English use or proficiency. These are location groupings, not personal language labels.

Figure 2 reads aggregate to decomposition to sensitivity: Fig. 2A contrasts core Anglophone with all other countries, Fig. 2B decomposes that aggregate into China, India, and other countries, and Fig. 2C reports the 2024-minus-2018 endpoint change under the 44 observed specifications and the denominator-missingness

bound. Aggregate PRI is recomputed from summed accepted and production credits and never averaged from group PRIs. The descriptive displays move to Fig. S3. Fig. S3A regroups the four primary groups into two language contexts: Anglophone context combines core Anglophone and India; non-Anglophone context combines China and other non-core. The accepted and production credits are each summed before recomputing shares and PRI; country or group PRIs are not averaged. These labels describe a regrouping of institutional locations, not an exhaustive classification of actual language use or personal proficiency. Fig. S3B shows selected country references.

Figure 2B decomposes the "all other countries" aggregate of Figure 2A into China, India, and other countries, so that the four primary groups appear at the level where the aggregate is not homogeneous. Core Anglophone is identical in both panels. Fig. S3B separately shows individual-country reference trajectories for the United States, South Korea, China, and India; these are descriptive illustrations, not a prespecified or statistically representative sample. All trajectories use the same primary country-credit and production-denominator rules, and the primary analytical groups are unchanged.

Supplementary Table S3 is included in **Additional file 2, Supplementary_Table_S3.csv**. It reports accepted-paper credits and shares, AI output credits and shares, and PRI for 2018 and 2024 for the 30 countries and regions with the largest observed AI output credits in 2018. The same countries and baseline ordering are retained at both endpoints, and shares use the complete observed annual country-credit totals. The same CSV is included in Additional file 2 alongside the full 210-country/region endpoint and 2018–2024 annual CSVs. These are descriptive expansions of the main-text Table 2 primary specification, with no new collection or model fitting.

S2.2 Two specification families: 12 country-API-aggregate and 32 work-level specifications

The archived dashboard contains 44 specifications in two distinct families, each evaluated for four country groups:

- **Country-API-aggregate family (12):** two country-mapping rules $\times$ two numerator counting rules $\times$ three production denominators. The denominators are AI-primary-subfield article/conference-paper records, AI-primary-subfield records of all types, and computer-science-primary-field article/conference-paper records.
- **Work-level family (32):** two country-mapping rules $\times$ two numerator counting rules $\times$ eight work-level denominator rules. The denominator rules are all-authorship paper-country full and fractional counting; authorship-fractional counting over all or country-covered authors; strict first-position full and fractional counting; and first-position full and fractional counting with a first-row fallback when position metadata are absent.

The primary specification belongs to the first family: `existing|conservative|top_full|ai_primary_peer_reviewed`. The internal string "peer_reviewed" labels a document-type filter, not an

independent audit of peer review. The two families are retained separately in the `specification_family` column of the archived dashboard (`existing_inspected_12spec` and `worklevel_32spec`). The phrase “32 work-level specifications” does not include the 12 country-API-aggregate specifications or redefine the primary denominator.

The audit is diagnostic rather than a model-averaging exercise. Specifications with different author positions or counting rules represent different estimands and are not pooled. In both families, endpoint signs agreed with the primary observed-data trajectories: China and other non-core increased; core Anglophone and India decreased.

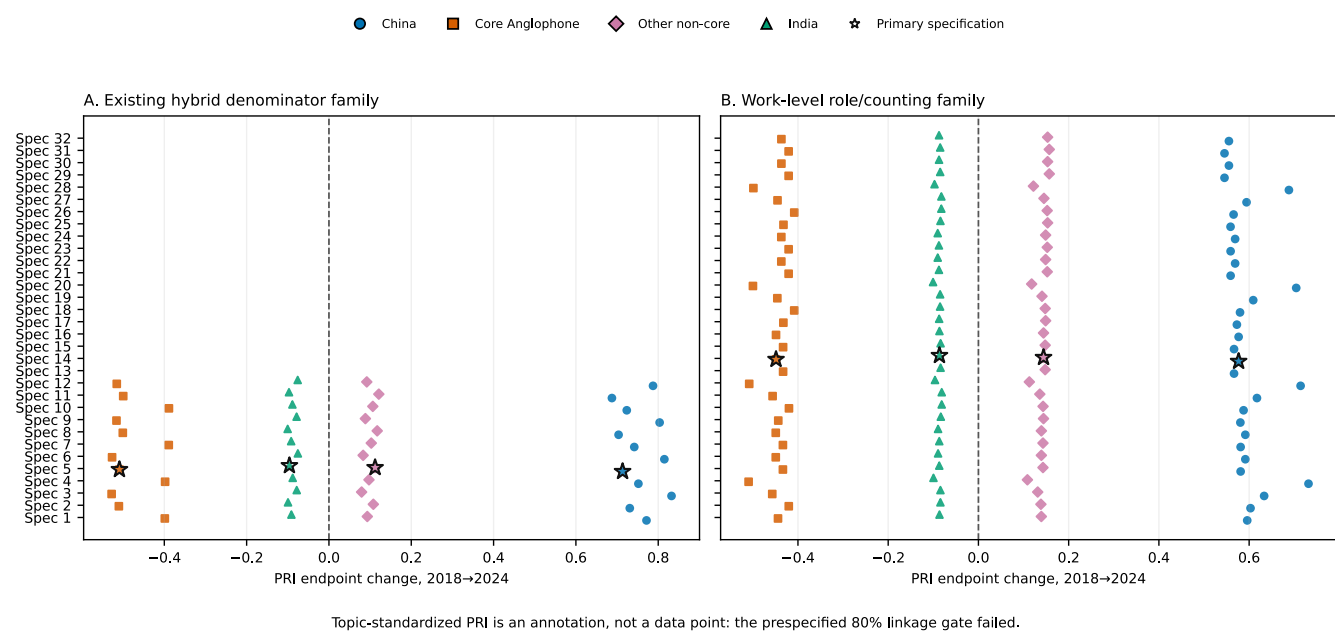


Fig. S1 PRI specification audit. Endpoint changes across the 12 country-API-aggregate and 32 work-level specifications. The highlighted specification belongs to the aggregate-based family and corresponds to the manuscript's primary construct. Complete values and family labels are in `table_s9_pri_specification_dashboard.csv`.

S2.3 Country-missingness bounds and their scope

Observed-data specification stability does not imply stability to unknown country allocation. The denominator analysis fixes the observed accepted-program share and varies the country composition of work-level records without strict first-position country evidence. It uses the conservative numerator mapping and the denominator `role_matched_first_position_fractional_strict`. Fractional and full numerator counting are examined separately. This is not the all-authorship country-aggregate denominator used for the primary PRI.

For year t , let N_t be the number of cached work-level records, O_t the number with observed strict first-position country evidence, and $M_t = N_t - O_t$. Let $C_{g,t}$ be group g 's observed fractional credit and $q_{g,t} \in [0, 1]$ its mean fractional credit per missing-country record. The completed production share is $(C_{g,t} + q_{g,t}M_t)/N_t$. Bounds for the 2024-minus-2018 PRI difference vary the two endpoint allocation parameters independently over their allowed ranges. They are allocation bounds, not sampling confidence intervals.

For China, the fractional-numerator specification has observed PRI 0.393582 in 2018 and 0.974733 in 2024, with change bounds $[+0.014359, +1.116463]$. Its increasing direction is identified within this allocation model. With the same strict fractional denominator but a full-count numerator, the observed PRI is 0.404628 and 0.971162, and the change bounds are $[-0.005006, +1.107522]$. The latter include zero, so an increasing direction is not identified throughout the allowed allocation range. Core Anglophone, other non-core, and India have change bounds spanning zero under both numerator counting rules. Main-text Table 3 displays all eight rows.

Break-even geometry under denominator missing-country model

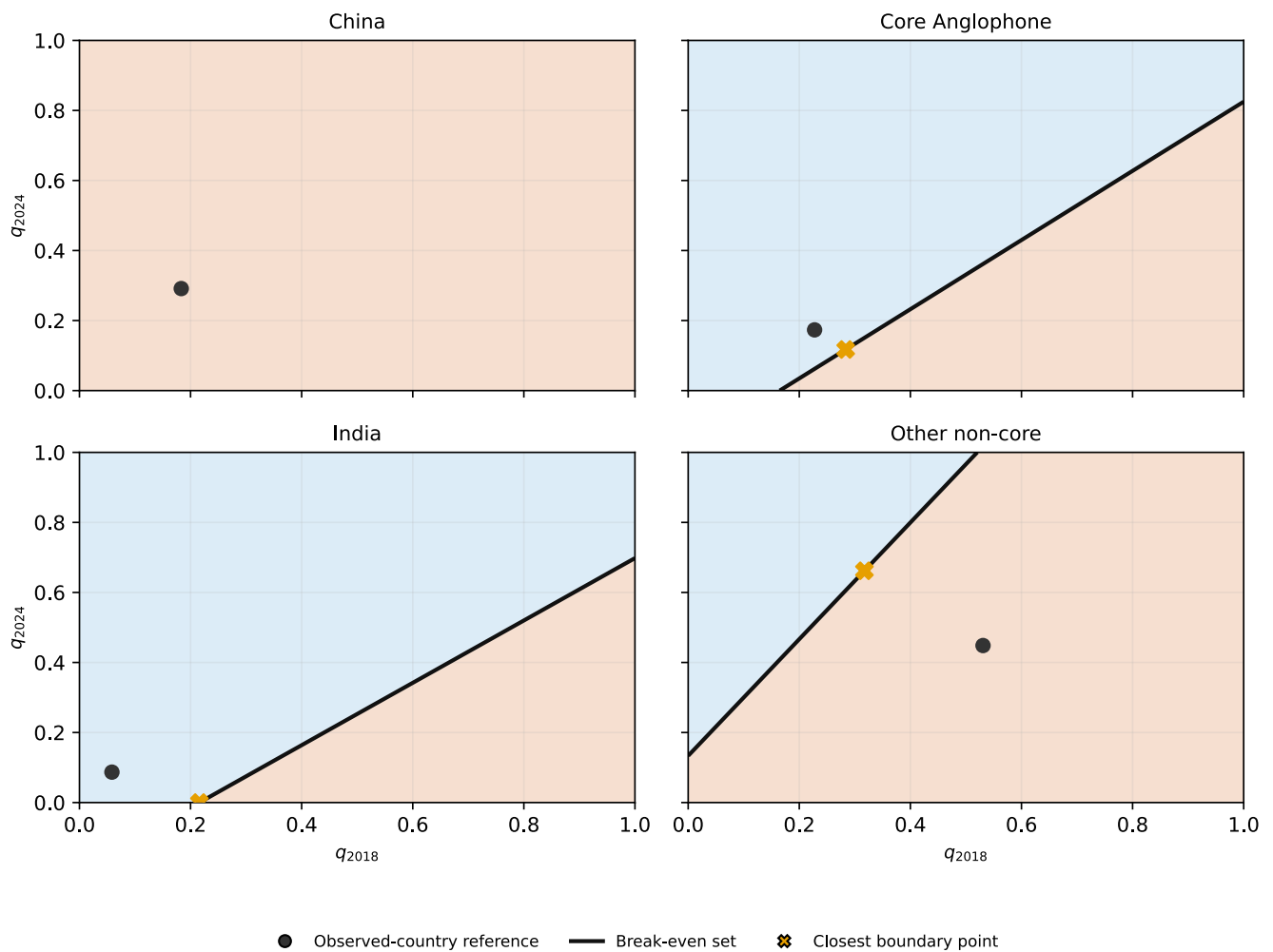


Fig. S2 Denominator-country missingness analysis, fractional-numerator case. The archived figure displays the fractional-numerator analysis with the strict first-position fractional denominator. China's sign identification in this figure must not be extended to the full-numerator analysis or to the primary all-authorship country-aggregate PRI. Both numerator counting rules, their bounds, and break-even parameters are reported in `table_s10_denominator_missingness_bounds.csv` and main-text Table 3.

The numerator analysis is separate: it holds the observed primary production shares fixed and allocates one total country credit per unresolved accepted paper. The observed numerator is evaluated under full and fractional counting, but the completion assumption remains one total credit per unresolved paper. For China under the full-count observed numerator, the zero-change boundary intersects the allocation square near $(q_{2018}, q_{2024}) = (0.868, 0)$ and $(1, 0.263)$. Its observed increase can therefore reverse for some allowed allocations. The fractional-numerator alternative is also not sign-identified throughout the allocation square. Complete values are in `table_s11_numerator_missingness_boundary.csv`.

The q in $[0, 1]$ range places almost no restriction on the country composition of the unobserved records, so it is a worst-case partial-identification stress test rather than a plausible-range interval. A restricted version confines the average credit assigned to an unobserved record to a band of half-width δ around the observed group share s , that is q in $[\max(0, s - \delta), \min(1, s + \delta)]$. $\delta = 0$ reproduces the observed endpoint change and $\delta = 1$ reproduces the worst-case bound above, both to nine decimal places. δ^* denotes the smallest δ at which the 2024-minus-2018 change bound first contains zero. Under the full-counting numerator δ^* is 0.056 for core Anglophone, 0.152 for India, 0.218 for other countries, and 0.693 for China; relative to each group's mean observed share these are 28%, 210%, 44%, and 292%. Under the fractional-counting numerator China retains its sign to $\delta = 1$, and the other three groups give 0.056, 0.157, and 0.213. Because δ is an absolute share perturbation, δ^* depends on both group size and PRI level and should not be read as a bare ranking. This analysis re-collects nothing: the observed shares, accepted shares, and annual missingness rates are reconstructed from the frozen S10 table, and the same annual missingness rates (0.353381 in 2018, 0.232077 in 2024) are recovered independently from all eight group-by-counting-rule rows. The full δ grid is in `table_s10b_restricted_missingness_bounds.csv`.

Neither the denominator-only nor the numerator-only analysis is a joint bound varying both sources of missingness simultaneously. Neither addresses unindexed works or field-classification error. The reported extreme allocations are possibilities under the stated model, not claims about the actual missing-country distribution.

S2.4 Primary four-group trajectories and descriptive regrouping

The main figure separates a broad accounting view from selected country trajectories. The four-group primary analysis is retained here without changing its group definitions, credits, denominator, or annual values. Figure 2A is a descriptive regrouping and does not replace this estimand.

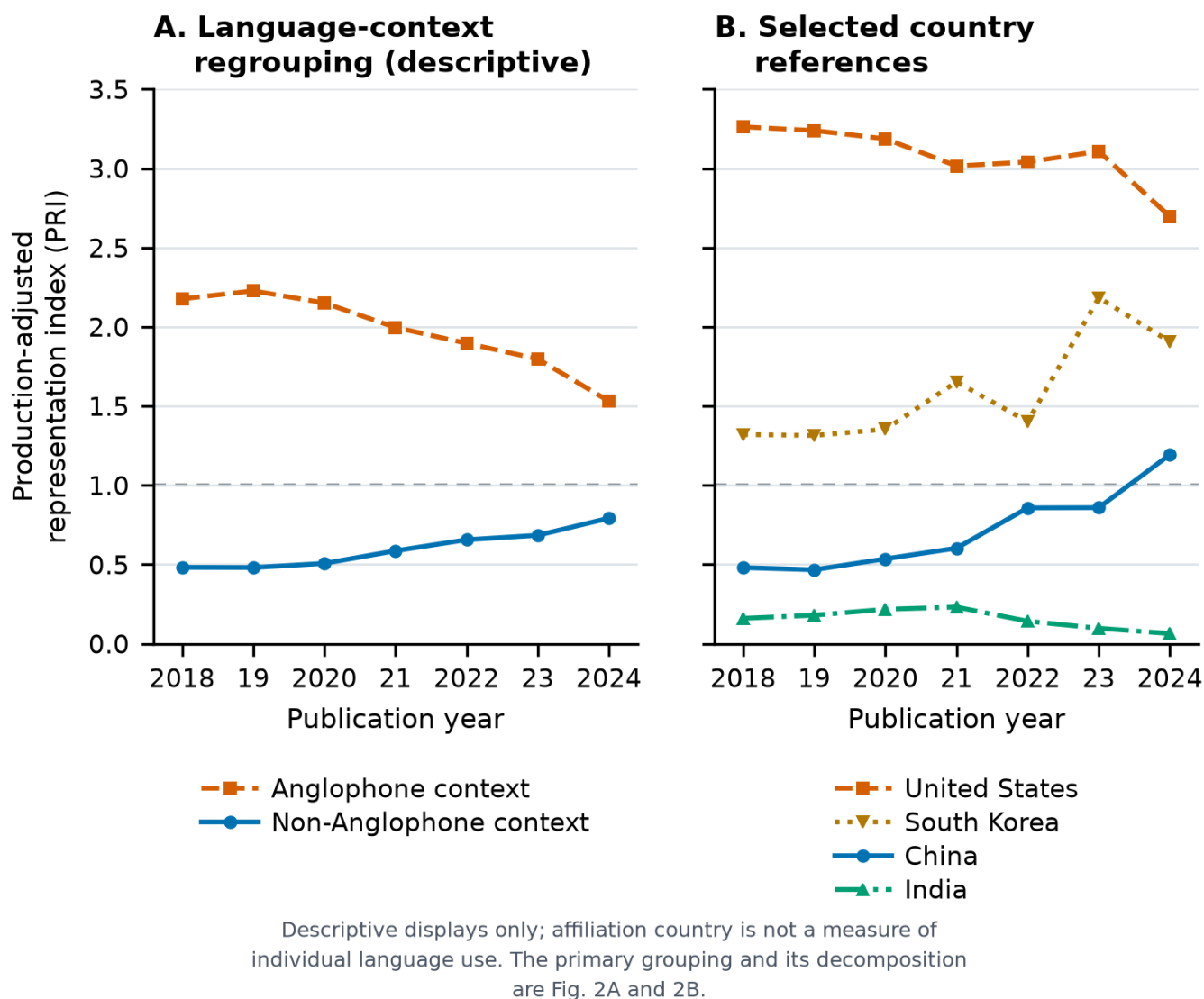


Fig. S3 Descriptive language-context regrouping and selected country references. Panel A regroups the four primary groups into an Anglophone context (core Anglophone and India) and a non-Anglophone context (China and other non-core); accepted and production credits are summed before PRI is recomputed, and group PRIs are never averaged. Affiliation country is not a measure of individual language use, so Panel A does not replace the primary estimand. Panel B shows the United States, South Korea, China, and India under the same credit and denominator rules: the United States is the largest core-Anglophone reference, South Korea sits outside the core-Anglophone set with PRI above one, China is the largest observed increase, and India is the lowest production-adjusted case. The dashed line indicates equal observed country shares. The primary grouping and its decomposition are main-text Fig. 2A and 2B, primary endpoints are main-text Table 2, and missingness bounds under a different denominator are in Table 3.

For Figure 2A, Anglophone context combines core Anglophone and India, and non-Anglophone context combines China and other non-core. Credit aggregation yields PRI 2.177 to 1.531 for Anglophone context and

0.482 to 0.792 for non-Anglophone context from 2018 to 2024. These observed-data descriptive directions must not inherit the sign-identification conclusions of a different missingness specification. Country-level labels cannot identify individual English proficiency.

S2.5 External production-denominator triangulation

OpenAlex missing-country bounds do not cover works absent from the index or time-varying country-specific coverage. We therefore compared the direction of the OpenAlex primary production-share series with Country AI Activity Metrics version 1.11.0 from ETO. ETO derives research counts from the Merged Academic Corpus, applies an independent AI classifier, and assigns a country one credit when at least one author organization on an article maps to that country. We used the AI-wide `field = All` rows for 2018–2024; every included row was marked materially complete by ETO.

The ETO country field contains individual countries or territories alongside regional and political groupings. Before calculating the denominator, we excluded 13 aggregate entities: ASEAN, Africa, Asia, EU, Europe, Five Eyes, Global Partnership on Artificial Intelligence, NATO, North America and the Caribbean, OECD, Oceania, Quad, and South and Central America. The remaining country/location rows define a full country-credit denominator analogous in direction, though not identical in construction, to the OpenAlex work-country denominator. As a denominator sanity check, the all-entity totals were 780,878 credits in 2018 and 1,533,194 in 2024, whereas the country-only totals used here were 191,092 and 423,018.

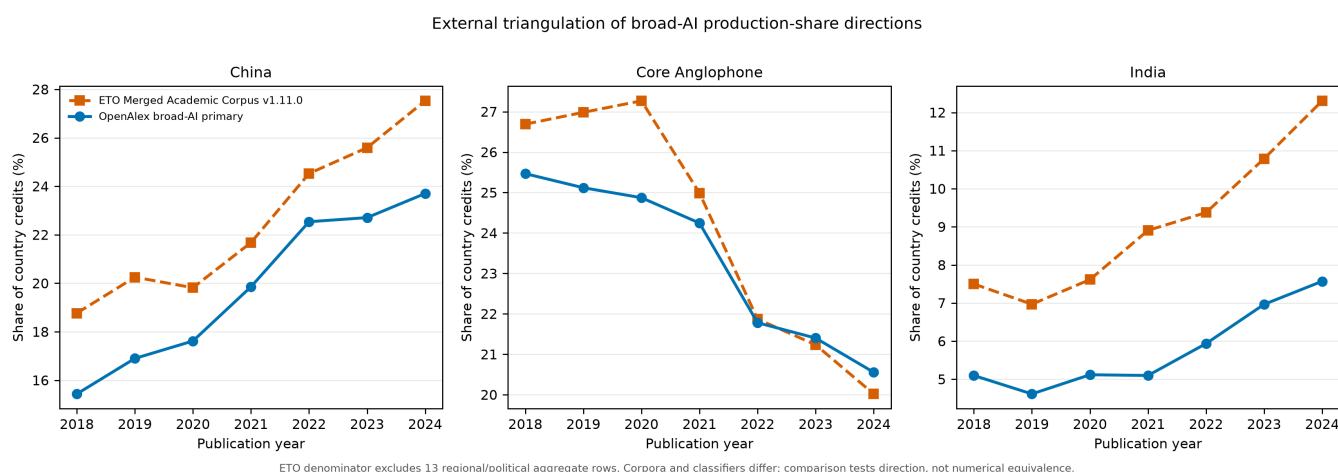


Fig. S4 External production-denominator triangulation. China’s country-credit share increases, the core-Anglophone share decreases, and India’s share increases in both the OpenAlex primary denominator and country-only ETO-MAC series. The comparison tests endpoint direction rather than level equivalence because corpus construction, AI classification, coverage, and counting rules differ.

Source	Group	Share 2018	Share 2024	Relative change	Direction
OpenAlex broad-AI primary	China	15.4%	23.7%	+53.5%	Increase
ETO-MAC v1.11.0, country-only	China	18.8%	27.5%	+46.7%	Increase
OpenAlex broad-AI primary	Core Anglophone	25.5%	20.6%	-19.3%	Decrease
ETO-MAC v1.11.0, country-only	Core Anglophone	26.7%	20.0%	-25.0%	Decrease
OpenAlex broad-AI primary	India	5.1%	7.6%	+48.5%	Increase
ETO-MAC v1.11.0, country-only	India	7.5%	12.3%	+64.1%	Increase

The three matching directions provide a source-sensitivity check on broad production-share change, not on accepted-program PRI itself. In particular, ETO-MAC corroborates the production-growth component of the India description but cannot make India's PRI decline identified under OpenAlex missing-country bounds. ETO documentation also cautions that its primarily English-language sources omit many Chinese-language publications and that some country-linkage errors remain. Annual values, exact endpoints, raw all-entity totals, aggregate exclusions, and the frozen denominator rule are in `table_s12_external_production_trajectories.csv`, `table_s13_external_production_endpoints.csv`, and `reviewer_additions_diagnostics.json`.

S3 Entrant definition and venue-universe asymmetry

Index events occur in ICML or NeurIPS. The five-cycle lookback and primary return universe include AAAI, ICLR, ICML, and NeurIPS. The broader history universe reduces two errors:

- an author with a recent AAAI or ICLR record being classified as focal-set-new at ICML or NeurIPS; and
- an entrant with a later AAAI or ICLR record being classified as having no focal-set reappearance.

The asymmetry is an observation-universe decision, not a claim that the four venues represent all AI. RQ3 sensitivities use ICLR/ICML/NeurIPS and ICML/NeurIPS-only return universes. The entrant label remains conditional on DBLP identity, four venues, and five publication cycles.

S4 Distance construction and calibration

S4.1 Portfolio construction

Primary prior portfolios use retrieved DBLP titles in the five calendar years before the index year. Index portfolios use the titles of all qualifying first-listed-author ICML and NeurIPS index papers for the entrant during the index year. Primary prior records are DBLP article and inproceedings entries, deduplicated by normalized title. Each SPECTER2 proximity-adapter title vector is L2-normalized; vectors are then averaged within each portfolio and L2-normalized again. The outcome is one minus cosine similarity between the two portfolio means.

Entrants without a retrieved prior title receive no primary distance. No maximum, minimum, or sample-mean value is imputed. Alternative measures are word TF-IDF, character TF-IDF, and SPECTER2 over expanded DBLP record types.

S4.2 Prespecified random-portfolio benchmark

The calibration sample was frozen before the full analysis and stratified over 2018, 2022, and 2024. Of 306 sampled entrants, 249 had an observed primary distance. For each focal entrant, the reference pool consisted of **other distance-observed pilot entrants in the same index-year stratum**. One hundred prior portfolios were sampled from this pool and paired with the focal entrant’s index portfolio.

For each entrant, let the own-portfolio distance be the observed title distance and let the randomized median distance be the median of 100 randomized distances. The calibration margin is

$$m_i = \tilde{d}_i^{rand} - d_i^{own}.$$

The own portfolio was closer than the randomized median for 73.5% of observed pilot entrants. The paired median margin was 0.026, versus an observed-distance IQR of 0.048. Word TF-IDF and SPECTER2 distances had Spearman correlation 0.692. All six frozen gates passed.

Table S2 Distance calibration gates

Metric	Value	Frozen criterion	Result
Pilot authors	306	Frozen stratified sample	Pass
Authors with observed distance	249	Coverage $\geq 70\%$	Pass
Finite distance rate	1.000	$\geq 99.9\%$	Pass

Metric	Value	Frozen criterion	Result
Distance IQR	0.048	≥ 0.02	Pass
Word TF-IDF Spearman correlation	0.692	≥ 0.15	Pass
Own portfolio closer share	73.5%	$\geq 55\%$	Pass
Paired median margin	0.026	> 0	Pass

Machine-readable source: table_s2_distance_calibration_gates.csv.

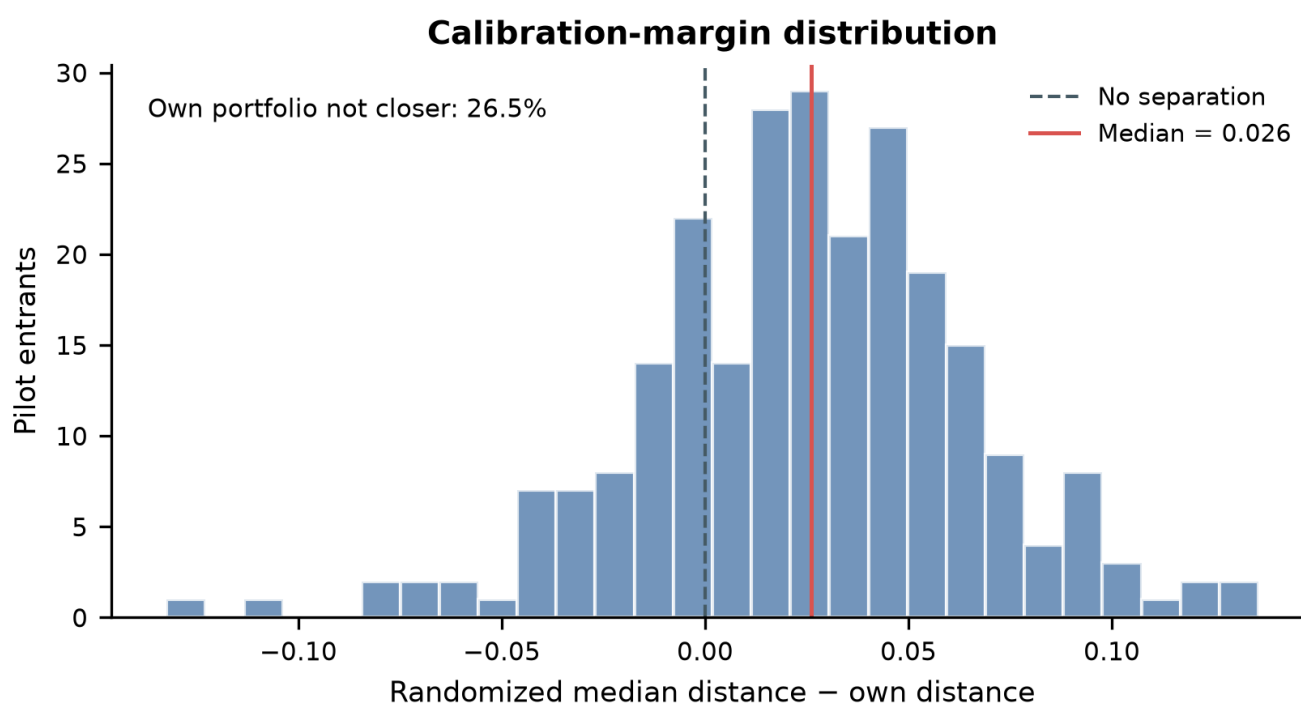


Fig. S5 Calibration-margin distribution. Positive values indicate that an entrant’s own prior portfolio was closer to the index portfolio than the median of 100 same-year randomized prior portfolios. The own portfolio was **not** closer for 26.5% of observed pilot entrants. This nontrivial tail limits individual-level interpretation: the benchmark establishes aggregate discrimination, not a true semantic ranking for every author.

S4.3 Distance observation

Primary distance was observed for 9,639 of 12,094 entrants (79.7%). Coverage increased from 74.3% in 2018 to 83.2% in 2024. Entrants without observed distance had mean team size 4.24 and experienced-coauthor share 0.859; those with distance had mean team size 4.68 and share 0.863. Their mean index years were 2021.52 and 2021.85, respectively.

The comparison does not establish missing at random. All RQ2, D1, and RQ3 distance claims are explicitly conditional on an observed prior-title portfolio. Detailed coverage by year, venue, team size, index output, and country-evidence status is in `table_s7_distance_coverage_by_dimension.csv`; the compact observed-versus-missing comparison is in `table_s3_distance_observation_missingness.csv`.

S4.4 Distance descriptions by country group

We linked each entrant’s index papers to first-listed-author affiliation-country evidence. Among 9,639 distance-observed entrants, 8,658 mapped unambiguously to one primary group. We excluded 795 with missing country evidence, 139 whose index-paper country list crossed primary groups, and 47 whose multiple index papers mapped to different groups.

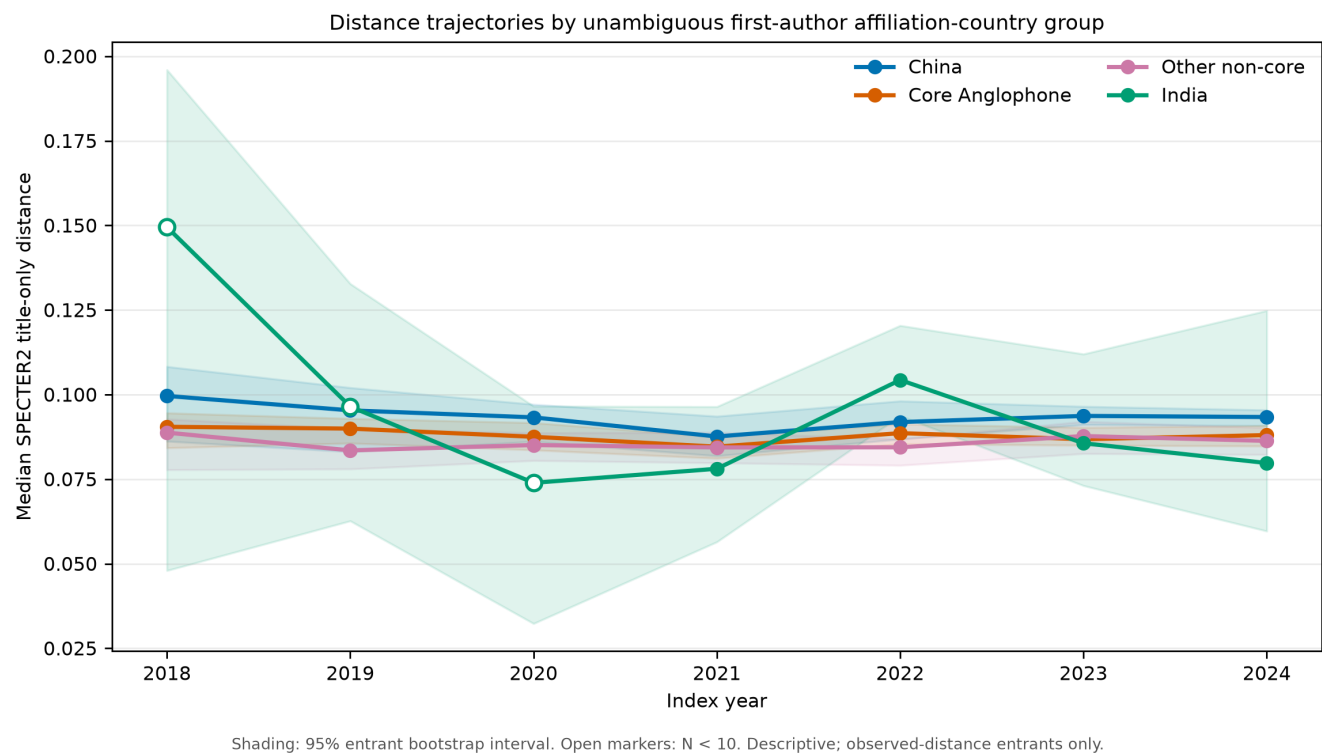


Fig. S6 Country-group distance trajectories. Annual median SPECTER2 title-only distance among distance-observed entrants with an unambiguous first-listed-author affiliation group. Shading gives 95% entrant-bootstrap intervals; open markers identify $N < 10$. These are accepted-cohort descriptions, not estimates of population-level national differences.

Group	Pooled N	Median	Q25–Q75
China	1,892	0.093	0.072–0.117
Core Anglophone	4,083	0.088	0.067–0.112

Group	Pooled N	Median	Q25–Q75
Other non-core	2,600	0.085	0.063–0.112
India	83	0.086	0.064–0.114

Annual medians were broadly stable within the larger groups. India had only 6–21 observed entrants per year, so its trajectory is noisy and should not be compared as if equally precise. These conditional summaries do not decompose or rule out compositional effects; accepted-only data cannot distinguish the distribution of researchers attempting entry from a distance range selected by the venues. Complete annual and pooled values are in `table_s14_country_group_distance_by_year.csv` and `table_s15_country_group_distance_pooled.csv`.

S4.5 Endpoint magnitude audit

An additional audit directly compares 2024 and 2018 within the distance-observed cohort. The median difference is -0.00130 , with a 95% entrant-bootstrap interval of $[-0.00494, +0.00157]$ from 2,000 within-year resamples (seed 20260827). The adjusted-mean difference is -0.00007 , with a paired HC3 coefficient-draw interval of $[-0.00365, +0.00344]$. The absolute median change is 2.8% of the pooled distance IQR and 5.0% of the pilot calibration margin; the annual median maximum-minus-minimum range is 0.00640.

These are descriptive magnitude comparisons. No equivalence margin was prespecified; an interval including zero does not establish absence of change or equivalence. This audit does not replace the primary annual trajectory or resolve selection into observed title histories.

S5 Diagnostic D1 sensitivity

D1 asks whether entrants at distance Q75 had a higher adjusted probability of an experienced coauthor than entrants at Q25. The primary standardized probability difference was +0.5 percentage points (95% interval -0.3 to $+1.4$; $N = 9,521$).

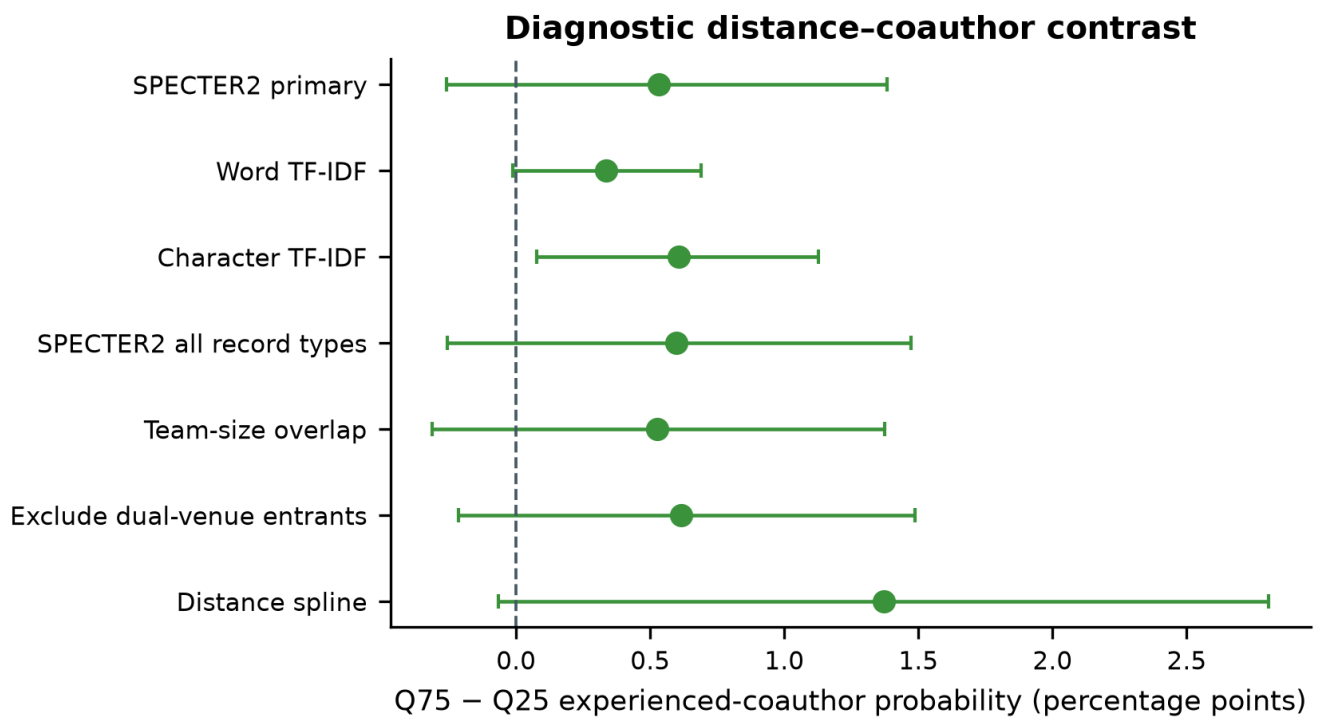


Fig. S7 Diagnostic distance-coauthor contrast. Average standardized experienced-coauthor probability difference between distance Q75 and Q25. Intervals use paired HC3 coefficient draws. The primary and alternative specifications are small in magnitude; several intervals include zero.

Alternative point estimates ranged from +0.3 points for word TF-IDF to +1.4 points for the distance spline. Character TF-IDF yielded +0.6 points with an interval above zero, but the magnitude remained small. The diagnostic conclusion is therefore based on scale as well as threshold crossing: no specification showed a large experienced-coauthor difference between quartiles. Complete estimates are in `table_s4_diagnostic_d1_sensitivity.csv`.

S6 Exact +2 outcome and censoring

S6.1 Complete risk set

The frozen bibliographic record ends in 2024. Exact +2 outcomes are fully observable for 2018–2022 entrants only, yielding 6,704 eligible entrants. Of these, 5,207 have observed distance; the mean index-team-size restriction of at least two leaves $N = 5,122$. This is a mean across qualifying first-listed index papers, not simply an exclusion of solo-only index events. Reappearance is observed at any author position.

Entrants from 2023 and 2024 are excluded from RQ3. They are never coded as having no reappearance. The D1 sample is larger (N = 9,521) because its outcome—experienced-coauthor status—is fixed at the index event and requires no follow-up.

S6.2 Why exact +2 is not within +2

Exact +2 was frozen as a fixed-horizon estimand before inspecting the outcome contrast. It gives every cohort the same publication-cycle position. A cumulative within-two-cycle outcome combines +1 and +2 opportunities, mixes immediate repeat publication with later reappearance, and changes the coauthor-continuity classification. It is a valid but different estimand and is not substituted for the primary outcome.

Publication cycle is not identical calendar exposure. “No record” means no observed accepted-paper record in the defined return universe at $t + 2$; it does not mean research exit.

S6.3 Within-two-cycle sensitivity

After freezing the primary result, we added a labeled sensitivity for any accepted focal-set record at +1 or +2. It retains the same 2018–2022 eligible cohorts, observed-distance and team-size restrictions, covariates, Q25/Q75 standardization, and 2,000 entrant bootstraps. Among 6,704 eligible entrants, 3,789 had at least one within-two-cycle record; the adjusted model uses N = 5,122.

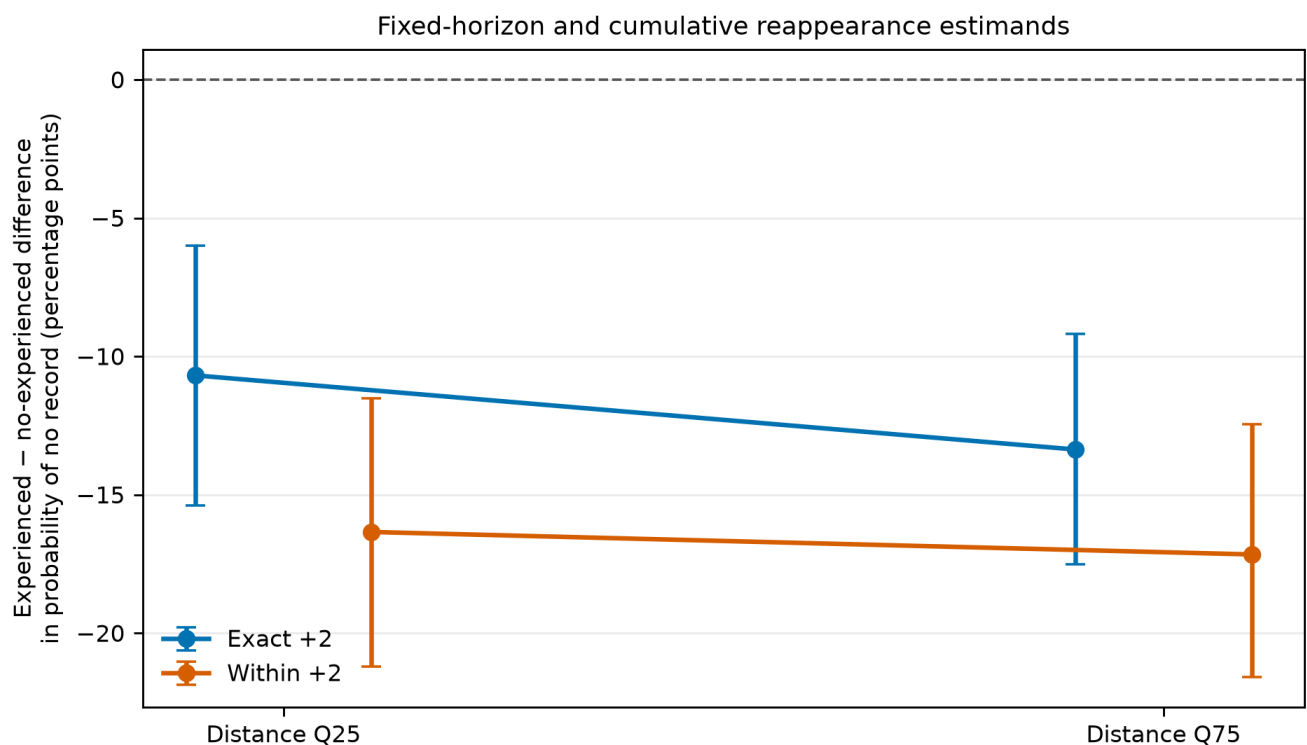


Fig. S8 Fixed-horizon and cumulative reappearance estimands. Experienced-minus-no-experienced differences in the adjusted probability of no record at distance Q25 and Q75. Exact +2 is the frozen primary estimand; within +2 is a labeled cumulative sensitivity.

Distance	Within +2 no-record contrast, percentage points	95% entrant-bootstrap interval
Q25	-16.3	-21.2 to -11.5
Q75	-17.2	-21.6 to -12.5
Q75 – Q25 difference-in-differences	-0.8	-5.2 to +3.9

The cumulative definition strengthens the overall experienced-coauthor association relative to exact +2, as expected when immediate +1 records are included. The interaction interval includes zero and does not establish a stronger association at high distance; it also does not establish equality between the two distance levels. Standardized probabilities and all pathway contrasts are in `table_s16_within_two_probabilities.csv` and `table_s17_within_two_contrasts.csv`.

S6.4 Model diagnostics

The primary multinomial model contained 15 design columns, had full rank, converged with finite parameters, contained all outcome categories, and completed all 2,000 requested entrant bootstraps. Primary SPECTER2 quartiles were Q25 = 0.065152 and Q75 = 0.112938. The primary bootstrap seed is 20261827 (base seed 20260827 plus 1000). Each replicate refits the model and standardizes over the resampled entrants while retaining original-sample distance scaling and quartiles. Complete primary and sensitivity model diagnostics are in `table_s8_rq3_model_diagnostics.csv`.

S7 Full RQ3 contrasts and interaction interpretation

The focal experienced-minus-no-experienced contrasts are reported in the manuscript. For completeness, the primary Q75-minus-Q25 difference-in-differences were:

Pathway	Difference-in-differences, percentage points	95% entrant-bootstrap interval
No exact +2 focal-set record	-2.7	-7.1 to +1.7
Continuity only	+1.5	-2.1 to +5.2
≥1 paper without an index coauthor	+1.1	-1.8 to +4.5

Point estimates use differences between unrounded contrasts from the original analysis sample. Interval limits are percentiles of the corresponding paired differences computed within each of the 2,000 entrant-bootstrap replicates. Every interval includes zero. Accordingly, these estimates do not establish a stronger experienced-coauthor association at high distance or equivalence between the two distance levels. The full file also reports within-coauthor distance contrasts and alternative distance and return-universe specifications: `table_s5_rq3_all_contrasts_and_sensitivities.csv`.

The sign-summary file, `table_s6_measurement_robustness_signs.csv`, is a compact audit aid. It is not a formal vote across models. Some alternative measurements target related rather than identical constructs, so sign disagreement qualifies interpretation instead of being averaged away.

S8 Reproducibility and provenance

S8.1 Frozen inputs included in the submission package

- `protocol_distance_pilot_v0.4.md`
- `config_distance_pilot_v0.4.json`
- `protocol_distance_v1.0.md`
- `config_distance_v1.0.json`
- `reporting_clarification_v1.0.md`
- `qa/asset_manifest.json`

The original package manifest records file paths, sizes, and hashes for its submission assets; it does not describe the present v05 revision. The v05 revision has a separate local QA record and provenance manifest in `EPJDS_v05_assets/revision_qa.json`. This reporting revision leaves the frozen CSVs, model estimates, and original submission assets unchanged.

S8.2 Source provenance and retained limitations

The code version cited in the main manuscript is the Beyond paper counts repository at Git commit `24f135413f203c56494a9c7b1388ce250f23b395`. The code uses the MIT license. Its offline frozen-asset validation was checked on Linux with Python 3.11. The software citation and project home page are provided in the main manuscript reference list. No separate study-archive DOI is claimed.

The exact retrieval date of the full DBLP dump and the immutable SPECTER2 model and adapter revisions were not recorded in the retained metadata. The original work-level OpenAlex intermediate outputs are not preserved; the associated 32 specifications remain available as frozen aggregate results. These limitations

are retained in the main manuscript Availability statement. Neither these clarification edits nor the country CSVs change the original analytical results.

S8.3 Main-table numbering and country-level supplementary files

Additional file 2 contains five CSV files corresponding to the original main analytical tables and twenty-one supplementary CSV datasets. Current main-text Table 3 relocates frozen results from supplementary CSV S10, and Table 5(c) relocates frozen results from supplementary CSV S5. The frozen filenames and numerical results are retained; the current six main-text tables do not imply six original main-table CSV files. The mapping is provided in `main_table_source_mapping.csv`.

Main-text table	Frozen CSV source
Table 1: populations	<code>table_1_analysis_populations.csv</code> ; <code>sample_flow_rq3.csv</code> for the intermediate distance-observed count
Table 2: primary geographic endpoints	<code>table_2_geographic_endpoints.csv</code>
Table 3(a–b): denominator-country worst-case bounds	<code>table_s10_denominator_missingness_bounds.csv</code>
Table 3(c): restricted $\pm\delta$ bounds	<code>table_s10b_restricted_missingness_bounds.csv</code>
Table 4: annual distance	<code>table_3_annual_distance.csv</code>
Table 5(a–b): D1 and RQ3 focal contrasts	<code>table_4_focal_probability_contrasts.csv</code>
Table 5(c): RQ3 contrast differences	<code>table_s5_rq3_all_contrasts_and_sensitivities.csv</code> , <code>semantic_primary</code>
Table 6: reporting bundle	<code>table_5_minimum_reporting_self_audit.csv</code> , with updated textual provenance references

The table mapping introduces no new model fit. The figure revision re-plots the same frozen country credits at two aggregation levels and adds no estimate. Additional file 2 includes the plot-input CSVs in a separate `figure_data` folder, without adding numbered supplementary tables.

The frozen original endpoint CSV retains the legacy field `direction_identified_under_denominator_missingness_bounds`. That field must not be read as sign identification for the primary all-authorship country API denominator. Current Table 2 omits it; current Table 3 and supplementary CSV S10 report the actual bounds by numerator rule under the separate strict first-position fractional denominator. The archived CSV is preserved byte-for-byte, and this interpretation note accompanies its release.

S9 Supplementary file index

File	Contents
table_s1_frozen_file_label_mapping.csv	Manuscript labels mapped to frozen internal file labels
table_s2_distance_calibration_gates.csv	Prespecified pilot validation gates
table_s3_distance_observation_missingness.csv	Compact observed-versus-missing distance comparison
table_s4_diagnostic_d1_sensitivity.csv	D1 estimates across distance and sample variants
table_s5_rq3_all_contrasts_and_sensitivities.csv	Full RQ3 contrasts and sensitivity estimates
table_s6_measurement_robustness_signs.csv	Compact sign audit
table_s7_distance_coverage_by_dimension.csv	Distance coverage audit
table_s8_rq3_model_diagnostics.csv	Convergence, rank, quartiles, and bootstrap diagnostics
table_s9_pri_specification_dashboard.csv	Geographic specification dashboard
table_s10_denominator_missingness_bounds.csv	Denominator-country boundary analysis
table_s10b_restricted_missingness_bounds.csv	Restricted $\pm\delta$ denominator-missingness bounds and δ^*
table_s11_numerator_missingness_boundary.csv	Numerator-country boundary analysis
table_s12_external_production_trajectories.csv	Annual OpenAlex and ETO-MAC production-share trajectories
table_s13_external_production_endpoints.csv	External production-share endpoint comparison
table_s14_country_group_distance_by_year.csv	Annual distance descriptions by country group
table_s15_country_group_distance_pooled.csv	Pooled distance descriptions by country group
table_s16_within_two_probabilities.csv	Standardized within-two-cycle pathway probabilities
table_s17_within_two_contrasts.csv	Within-two-cycle pathway contrasts
figure_s1_pri_specification_audit.pdf/svg	PRI specification audit
figure_s2_denominator_missingness_bounds.pdf/ svg	Denominator-country missingness bounds
figure_s3_language_context_and_country_reference s.pdf/png/svg	Descriptive language-context regrouping and individual-country reference trajectories

File	Contents
figure_s4_external_production_triangulation.pdf/png	Independent production-denominator direction check
figure_s5_calibration_margin.pdf/png	Random-portfolio margin distribution
figure_s6_country_group_distance.pdf/png	Conditional distance trajectories by country group
figure_s7_diagnostic_distance_coauthor.pdf/png	Diagnostic D1 forest plot
figure_s8_within_two_sensitivity.pdf/png	Exact-versus-cumulative reappearance comparison

Figure-data files in Additional file 2 are `figure_data/figure_2_panelA_aggregate_data.csv`, `figure_data/figure_2_panelB_four_group_data.csv`, `figure_data/figure_2_panelC_sensitivity_data.csv`, and `figure_data/figure_s3_plot_data.csv`.

Additional country-level files:

File	Contents
Supplementary_Table_S3.csv	Additional file 2: 30-country/region comparison for 2018 and 2024; also included in Additional file 2
supplement_table2_all_countries_endpoints.csv	All 210 observed country/region codes at the 2018 and 2024 endpoints
supplement_table2_all_countries_2018_2024.csv	Full annual series, 1,470 country-year rows